



F1 Data Analysis and Tactical Insights:

Exploring Formula 1 Race Performance Strategies

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ABSTRACT

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The purpose of the study was to investigate the complexities of analyzing Formula 1, and to examine the methods and strategies employed by various teams. By utilizing both the Fast F1 and Anaconda frameworks, pre-processed data was accessed and analyzed to reveal valuable insights into race dynamics and the implementation of strategies. The study encompassed two case studies to accomplish the intended objective.

Two case studies were carried out, one concentrating on telemetry analysis and the other on tire performance. The objective was to examine the influence of analysis metrics on race performance and strategies. The results indicate that although the analysis may not provide precise predictions, it does have a notable impact on distinguishing various team strategies.

The study underscores the need for practitioners in Formula 1 data analytics to gain a deeper understanding of the technologies utilized and improve analysis planning to enhance the quality of the visual representations of race dynamics and performance metrics. This thesis can contribute to the ongoing evolution of Formula 1 data analytics, offering valuable insights for teams, researchers, and enthusiasts seeking to enhance their understanding of race dynamics and optimize performance on the track.

Keywords: data analysis, Anaconda, Python, F1 Technical Analysis, Weather,

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GLOSSARY or ABBREVIATIONS AND TERMS (choose one or other)

API	Application Programming Interface
DRS	Drag Reduction System
F1	Formula 1
FIA	Federation Internationale de l'Automobile
Fig	Figure
GP	Grand Prix
IDE	Integrated Development Environment
KM	Kilometre
KM/H	Kilometre per hour
TAMK	Tampere University of Applied Sciences

1 INTRODUCTION

This thesis utilizes the Fast F1 framework as the primary data source for analyzing Formula 1 races, encompassing various aspects such as race performance strategies, degradation, and driver performance. In addition to telemetry data, the framework incorporates supplementary information to provide comprehensive insights into race dynamics. As races progress, real-time data is extracted and stored, facilitating in-depth analysis facilitated by API endpoints. The Fast F1 framework offers significant benefits to data engineers, providing convenient and cost-effective access to a wealth of race-related data and insights.

This data can be used to conjure up tactics and unearth opposing teams' tactics, it can be used to understand racers' habits, patterns, and strengths. Moreover, if processed correctly can be used to make correct onsite decisions thus affecting the outlook of a race.

Examining several sources, it seems there is an abundance of race analysis available however, few catered to the significance and anatomy of the types of techniques utilized in Formula 1 data analysis. Such as Telemetry analysis, Track analysis, Tyre degradation, circuit analysis and so much more. (F1 Racing Teams, 2024)

By examining different case studies, what is the impact of different race analysis techniques and how do they affect race strategies?

The thesis will also look at different test cases, and specific events, process, analyze, and visualize data to produce meaningful outcomes. Including but not limited to examining the methods, data, and sources used.

2 BACKGROUND AND PURPOSE

2.1 Background

In 1977, American physicist Dr. Karl Kempf pioneered the use of telemetry by equipping race cars with sensors to record various data points during races. This data, now integral to motorsports, includes speed, throttle position, brake pressure, tire conditions, suspension dynamics, and engine performance. (Unveiling the key data points in formula 1 racing)



FIGURE 1 Karl Kempf installing different sensors (Karl Kempf, 1977)

Formula 1 is a popular world sport where innovative teams come together to compete in their engineering to achieve success around the world. However, it's not the only reason to pay attention to it. This sport is primarily based on teams who come up with their own cars, designed from scratch to compete and dominate on the race track.

Formula One racing takes drivers and cars through a global tour of varying conditions, including weather, road surfaces, and human factors. The competition demands cars to be both agile and durable, enduring distances between 260 km and 360 km while reaching speeds of up to 360 km/h and navigating rapid speed changes.

Achieving success requires powerful yet lightweight vehicles capable of handling high speeds and sharp corners. Engineering prowess and financial investment are essential to crafting these lightning-fast tanks, with the ultimate goal being to cross the finish line after completing a predetermined number of laps.



FIGURE 2. RB20, Red Bull Racing Car with Driver Max Verstappen. (F1 Racing Teams, 2024)

FIGURE 2 shows Max Verstappen cornering at high speeds, exemplifying the intense demands placed on Formula One (F1) drivers and their vehicles. It showcases the agility and precision required of drivers, as well as the importance of vehicle durability and engineering excellence. This highlights Formula One racing as a thrilling and dynamic experience that captivates a worldwide audience.

2.2 Case Studies

2.2.1 Belgium GP 2023

This case study delves into telemetry data analysis from the Belgian Grand Prix 2023, aiming to extract insights into the factors influencing race outcomes. Intriguing patterns within the Red Bull Racing team were uncovered, particularly their consistent throttle disengagement at specific corners, prompting scrutiny of their strategic decisions. Initial theories regarding weather, track conditions, and driver competence were ruled out, leading to investigation into maintenance issues and radio communications, ultimately revealing a calculated response to navigate the complexities of the race track.

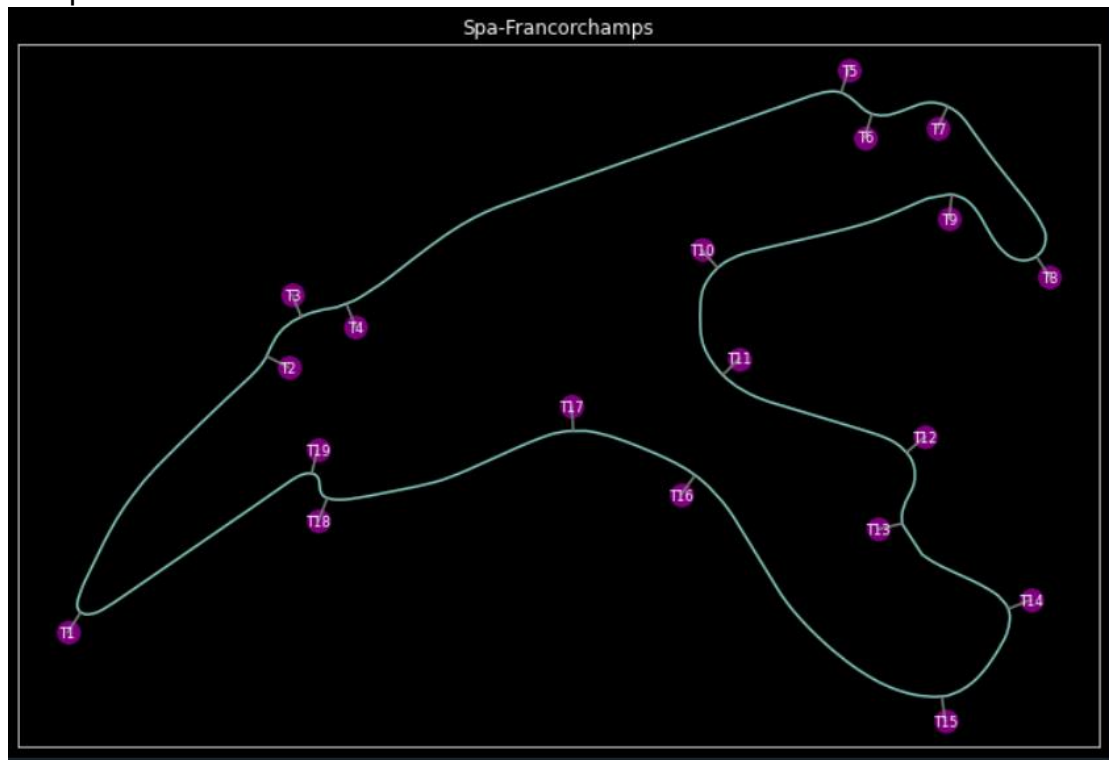


FIGURE 3: Belgian GP Circuit Spa-Francorchamps.

The culmination of this analysis suggests that a combination of factors, including regulations governing ride heights and slip deterioration effects, influenced Red Bull's cautious approach at specific corners, notably Eau Rouge.

These findings underscore Red Bull's tactical acumen and highlight the delicate balance between compliance with regulations and strategic advantage on the Belgian Grand Prix track.

2.2.2 Dutch GP 2023

The case study of the Dutch Grand Prix 2023 delves into tire performance through data analytics. With a focus on thorough analysis, this study aims to unravel the impact of tire behavior on race strategies. By examining data specific to this event, insights into the performance of different tire compounds and their impact on race outcomes will be uncovered.

The Dutch Grand Prix circuit, known for its fast corners and varying track conditions, presents an ideal setting to scrutinize tire performance and its role in determining race success. Shifting the analytical lens toward tire performance, this study takes a comprehensive approach to understanding its implications for the Dutch Grand Prix 2023.

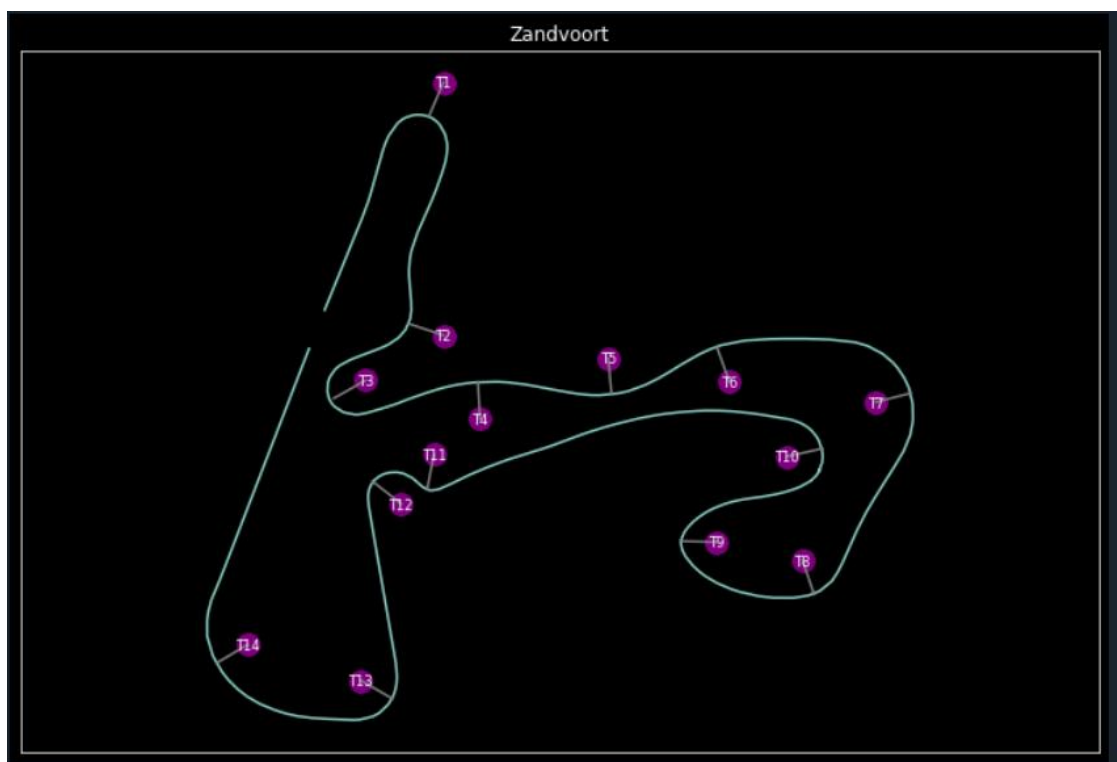


FIGURE 4: Dutch GP circuit Zandvoort.

Unlike previous investigations, which targeted specific incidents, this analysis adopts a broader perspective. By examining tire performance patterns across select laps and compounds, a nuanced understanding of race dynamics emerges. Through metrics such as LapTime and degradation, the study aims to decipher the complex interplay between driver performance, tire behavior, and race strategy on the challenging circuit of the Dutch Grand Prix.

2.3 Project Objective

In Data analytics in Formula One, measuring performance on the track is very important, it gives an insight into potential problems and successful aspects that can positively affect the performance of a race car. (Segers, 2014)

All Formula One cars feature logging systems that meticulously extract complex data, offering detailed insights into various aspects of the car/driver combination. This data provides an understanding of performance across different sections of the car. It elucidates how external factors such as track conditions, weather, and aerodynamics influence overall performance. (Segers, 2014)

In this thesis, the focus will be on a viewing two formula one race as case studies to analyze within the realm of Formula One racing, delving into the intricacies of data analytics as applied to this high-stakes sport. This case studies will revolve around a specific team or set of teams, exploring their utilization of data analytics to enhance performance, mitigate errors, and outmaneuver competitors. By closely examining the strategies and techniques employed, aiming to gain insights into the evolving landscape of data-driven decision-making in Formula One. (Segers, 2014)

Analyzing how teams leverage data to optimize various facets of their operations, ranging from vehicle performance and race strategy to driver management and resource allocation.(Segers, 2014) Exploring the competitive dynamics within the sport, scrutinizing how teams harness data analytics to gain a competitive edge over their adversaries. By dissecting real-world examples and success stories, to elucidate the profound impact of data analytics on Formula One racing and its implications for the broader realm of sports analytics.

3 REQUIREMENTS AND TECHNOLOGIES

This thesis aims to dive deep into engineering and tactical analysis of Formula 1 races. Formula 1 dates back to the European Grand Prix championships of the 1920s. A key definition of an old and cultured competition with high-level competition. (History of Formula One, 2024).

To manage aspects of this project, one must take into account the requirements and desired outcomes. Additionally, consider employing efficient and adaptable technologies that can facilitate the process.

Another aspect one needs to consider is the knowledge of F1 mechanics and tactical insights into the race. Otherwise, simply attempting to engage the project would be an impossible task to understand and develop techniques.

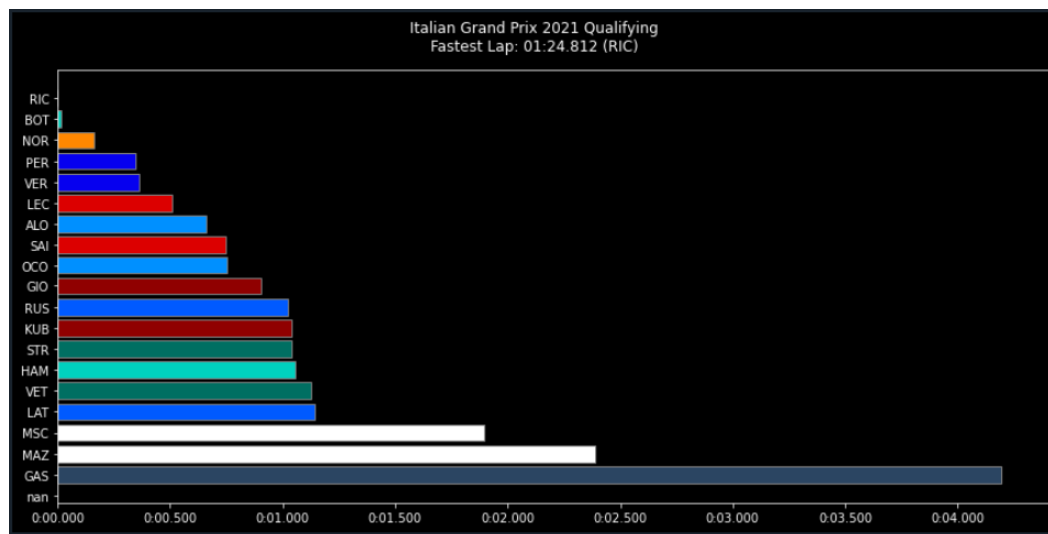


FIGURE 5. A graph of the Italian GP, showing the fastest laps in the race. (Baraka, 2024)

FIGURE 5 shows a list of the quickest laps in the Italian GP compared to the fastest lap achieved in the race by Driver Daniel Ricciardo (RIC) from the RB F1 team. To an individual without any F1 knowledge, the concepts of Laptimes, Delta Laptimes, and even driver names cannot convey concrete information.

FIGURE 5 is a simple overview to explain the overall racing outcomes. More complex would be discussed in core detail in the chapters to come.

3.1 Requirements

The following section shows the requirements, both functional and non-functional, for conducting a productive data analysis endeavor. From the acquisition and cleansing of the dataset to the execution of exploratory analysis and the selection of suitable visualization tools, each stage is methodically carried out to guarantee precise insights and proficient communication of findings. Furthermore, the iterative refinement process and the assimilation of contemporary methodologies amplify the intricacy and depth of the analysis. (Satrajit Ghosh, 2021)

3.1.1 Functional Requirements

These requirements outline essential functionalities serving specific purposes in data analysis.

- The analysis should feature clearly defined graph limits for accuracy.
- The analysis data should be plotted in X-Y graphs to demonstrate relationships effectively.
- The analysis should incorporate mathematically processed data for robust insights.
- The analysis should include comprehensive track mapping for context.
- The analysis should utilize telemetry and weather information extensively for thorough examination.
- The analysis should provide statistical data per lap or session for detailed analysis.(Segers, 2014)

3.1.2 Non-Functional Requirements

The following are the non-functional requirements essential for successful data analysis. These parameters ensure reliability, scalability, and efficiency throughout the analysis process. (Segers, 2014)

- The analysis should be readily comprehensible for users.
- The analysis should be sufficiently detailed for easy navigation.
- The analysis should include precise cursor mapping features like cursors, set markers, and distance measurement.

- The analysis should feature clear and coherent titles for clarity.
- The analysis should be thoroughly documented for reference.
- The analysis should be visually appealing to engage users effectively.

3.2 Development Technologies.

This segment offers insights into the technological landscape, detailing the platforms, programming languages, libraries, and other resources utilized to bring the project to fruition. Through this exploration, readers gain a comprehensive understanding of the technological foundation underpinning the project's execution and success.

3.2.1 Anaconda (Spyder IDE).

The analysis was done on the Anaconda distribution platform. It is a comprehensive open-source platform that plays a critical role in the management, development, and deployment of data science and machine learning projects. Data source is compatible with python and R languages. It offers an abundant variety of libraries, frameworks and tools essential for data science and machine learning tasks. (Unleash ai innovation and Value, 2024)

3.2.2 Python

Python was opted for due to its versatility, readability, and extensive ecosystems available. Python also offers simplicity and an abundance of libraries such as pandas, numpy, matplotlib, and tensorflow. The data source is a Python-based data source, as such python is a no-brainer. (Unleash ai innovation and Value, 2024)

3.2.3 Fast F1

Fast F1 is a python framework that gives access F1 telemetry, lap timing, position, car data, weather events, schedules and session results data. This python extension offers versatility, simplicity, and vast ecosystems of support online. Data was consistent and accurate with actual events. There were other options, however, they lacked depth and support. As such Fast F1 was utilized. (Fastf1, 2024)

3.2.4 Matplotlib

Matplotlib is a widely used Python plotting library, that facilitates the creation of static, animated, and interactive visualizations. With its user-friendly interface, Matplotlib is instrumental in presenting complex data insights through clear and compelling graphics. (Matplotlib 3.8.3 documentation, 2024)

3.2.5 NumPy

NumPy is a fundamental numerical computing library in Python, supporting large, multi-dimensional arrays and matrices, along with high-level mathematical functions. Its efficiency in numerical operations makes it indispensable for scientific computing, forming the foundation for various data science libraries. (Ecosystem, 2024)

3.2.6 Pandas

Pandas, a crucial data manipulation library, streamlines structured data handling with Series and DataFrame. Essential for efficient manipulation, cleaning, and analysis, it facilitates streamlined exploration and processing of real-world datasets for data scientists and analysts. (Pandas, 2024)

4 TECHNICAL IMPLEMENTATION

The following sections detail the technical implementation process.

4.1 Analysis Planning

Before delving into data analysis, it was essential to lay the groundwork through meticulous planning. This planning phase served as the foundation for the entire analytical process, guiding the direction of the investigation and ensuring that the efforts were focused and effective. During this stage, an overview of the dataset was conducted, gaining a comprehensive understanding of its structure, content, and potential insights it might hold.

Additionally, a specific type of analysis metric was identified that aligned with the objectives and requirements. By selecting an appropriate metric, the analytical approach could be tailored to address the key questions and challenges posed by the dataset. To illustrate this process, an example scenario was created that encompassed both an overview of the dataset and the application of a corresponding analysis metric.

Through this strategic planning phase, a solid framework was set for the data analysis endeavors, laying the groundwork for meaningful insights and informed decision-making.

4.2 Configuration and Installations

Before initiating the analysis, it is imperative to install and configure the required packages to unlock the necessary functionalities. These packages serve as the backbone of our analytical toolkit, providing access to a wide range of tools and capabilities essential for data exploration and manipulation.

The following are the steps to configure our environment,

- Step one, installing anaconda ecosystem which comes with the preferred Spyder IDE. Which grants access python language with built in dependencies like Pandas, Numpy as well as live graph production and display.

- Step two, Install Fast F1, external package dependencies that will be main source of data.
- Step three, installing assistant dependency packages that will elevate the main analysis. Installing matplotlib.

Note, when Fast F1 was installed, by default it installed more dependencies such as Scipy, contourpy and so forth.

4.3 Session Loading

With all dependencies successfully installed and verified, the next crucial step is to validate the functionality of FastF1. This entails importing the FastF1 library into the main file and initializing a session object. This session acts as a pivotal component, serving as the gateway to accessing and extracting essential data from the Formula 1 race.

Loading the session with essential parameters like the year, location, and race establishes a foundational framework crucial for subsequent analysis. This initialization process is fundamental, paving the way for the comprehensive exploration and interpretation of Formula 1 telemetry and race data.

With Python's flexibility in handling datasets within single variables and its array of built-in functions, the subsequent stages involve creating variables and manipulating them to access various information effortlessly.

5 CASE STUDY 1 ANALYSIS

Before conducting an analysis on any topic, it is customary to look at race overview. Analysing full grid performance and while also looking out for anomalies. Case study 1 details a Formula One race held at the iconic Spa-Francorchamps circuit in Belgium, located in Stavelot, Wallonia.

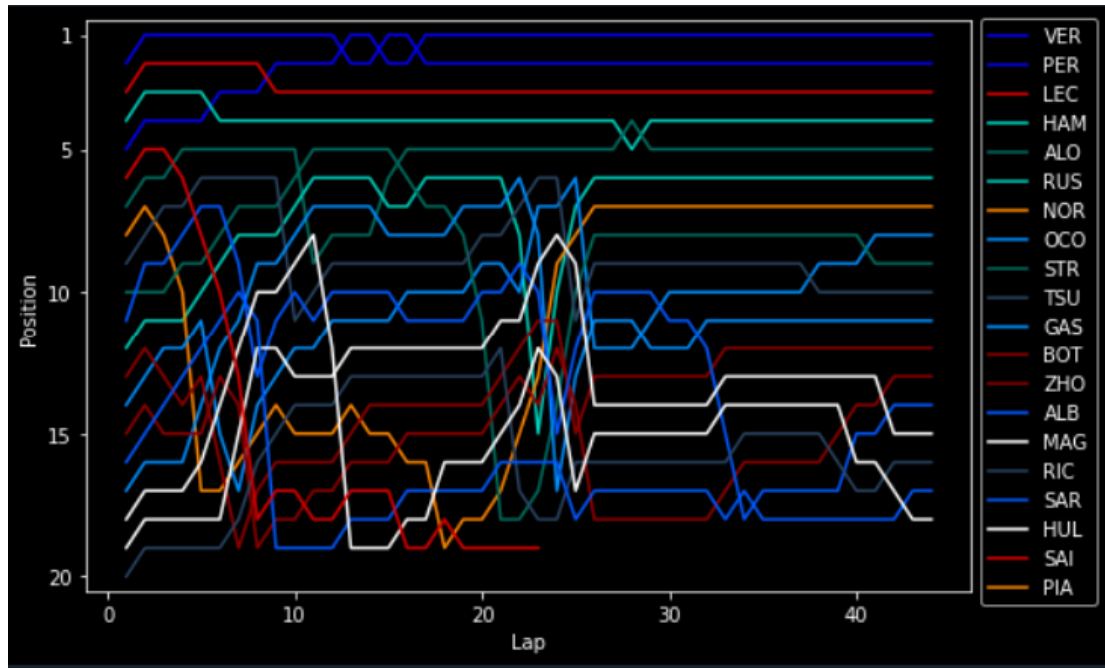


FIGURE 6. A Graph of Position Vs Lap Number. 1 (Baraka, 2024).

In perspective, at the end of this race Max Verstappen takes pole position, Lewis Hamilton takes position four. This overview was designed to feature each driver and specify their respective color to that of the team. Figure 6 represents the constant shifts and overtakes happening across the board.

Code Implementation

After loading the session and accessing the data stored in an array, it's time to begin manipulating this dataset to create the comprehensive overview mentioned earlier. Fortunately, Fast F1 provides extensive documentation at <https://docs.fastf1.dev/examples/index.html>. This documentation includes all the relevant built-in functions for manipulating the datasets, along with code examples demonstrating specific usage scenarios to achieve desired outcomes.

The following details relevant steps taken to achieve the overview,

- Two variables were created, one contained an array with drivers names loaded from “drivers = session.drivers” and the other contains lap-data loaded from “lap-data = session.laps”
- Then, in a for loop, each drivers laps is extracted using the drivers array as a parameter. This is achieved using a built in function, “lap-data.pick_drivers(drivers)”. Now, this resulting dataset contains all the columns and subsequent data that will be plotted using matplotlib.
- Fast f1 contains color codes as well that are labelled by the drivers names and teams names. Thus, “team_colors = fastf1.plotting.team_color(team)” utilizing the plotting function. One can finally have, multiple plots of data but visually appealing to specific teams.

However, this single illustration may not provide sufficient meaningful information. To gain a clearer understanding, it is customary to utilize multiple figures or graphs in tandem. This practice is prevalent throughout Formula 1 analysis, as the insights derived from a single graph are often limited in providing a comprehensive understanding of the data.

So, this time to get a better view, a pace variation graph in the fastest laps was implemented.

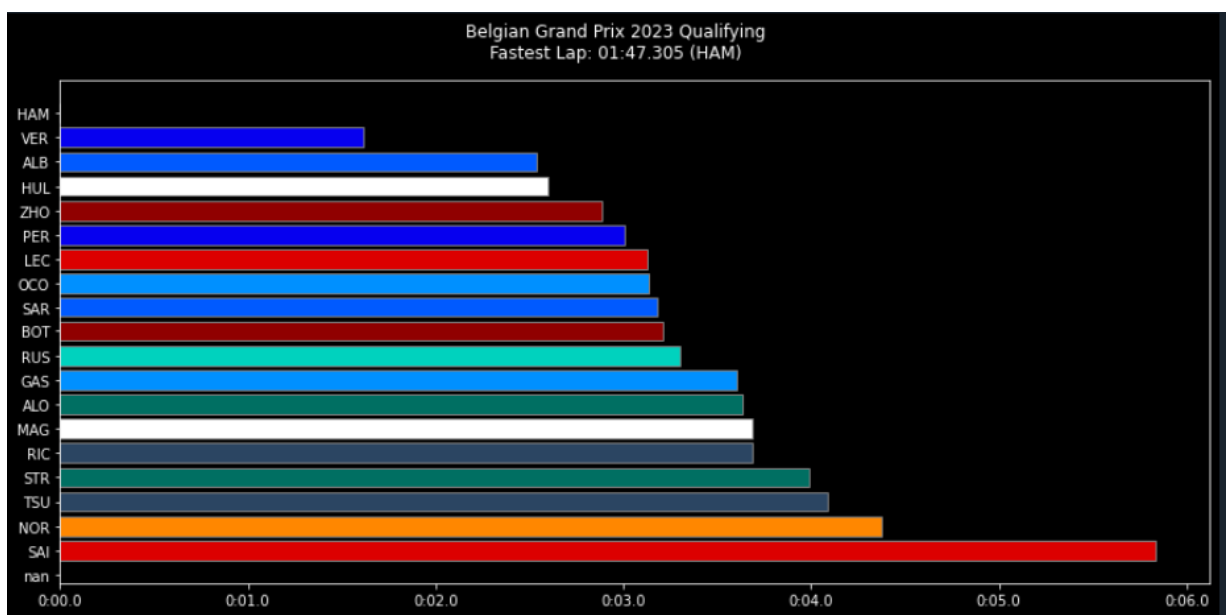


FIGURE 7: Drivers Pace lap in relation to the fastest lap (Baraka, 2024).

To achieve FIGURE 7, the relevant steps include,

- Initial variables were utilized similarly as Figure 6.
- In a for loop, to extract the fastest lap of each driver, using the method `"lap-data.pick_drivers(drivers).pick_fastest"`, and stored in an external array.
- The fastest lap was selected and stored as `"Pole_lap"`. Subsequently, the remaining laps were sorted from fastest to slowest.
- Each lap was then subtracted from the `"Pole_lap"`, and this value was plotted against the driver's index.

Now, Figure 7 presents a compelling insight when juxtaposed with Figure 6. While Figure 6 illustrates Verstappen's clear lead in the race, Figure 7 reveals that Hamilton significantly outperforms the rest of the drivers by a considerable margin.

This raised questions: What led to this shift? Why is Hamilton's performance notably different? To delve deeper into this intriguing anomaly, the subsequent method employed for investigation involved telemetry analysis.

5.1 Data Validation

Data validation as directed by Fast F1 documentation, is a critical step in ensuring the accuracy and reliability of the data used in this thesis. Two primary methods were employed to validate the integrity of the data: comparing calculated timestamps with known accurate values and checking the consistency of the data to ensure it aligns with expected patterns. (How to perform calculations accurately, 2023)

The first method involved using known accurate values, such as lap timing measured separately and known to be accurate to one millisecond. By calculating the start time of each lap and subtracting the beginnings of two subsequent laps, lap times were determined. These calculated lap times were then compared against the known accurate lap timing values. Any discrepancies between the calculated and known lap times would indicate potential errors or inaccuracies in the calculated timestamps.

Additionally, this validation method helped identify and mitigate any fixed offsets that might affect all timestamps uniformly, ensuring that random errors or general inaccuracies were excluded from the analysis. (How to perform calculations accurately, 2023)

The second method focused on checking the logical consistency of the data. For instance, when determining the start of a lap, the position of the car at that time was also calculated. It was expected that every time a lap started, the car would be crossing over the finish line, as the finish line is a static reference point that does not change. Any deviation from this expected behaviour would suggest errors or inconsistencies in the data. By systematically examining the relationship between timestamps and car positions, potential errors were identified and addressed, further enhancing the reliability of the telemetry data used in this thesis. (How to perform calculations accurately, 2023)

Overall, the combination of these validation methods provided a robust framework for ensuring the accuracy and integrity of the telemetry data, laying a solid foundation for the subsequent analysis and conclusions drawn in this thesis.

5.2 Telemetry Analysis.

In this section, after loading the session, it's decided what type of telemetry metrics would be useful. As discussed in previous chapters, telemetry consists of speed, throttle, DRS, position, suspension and engine performance. (Unveiling the key data points in formula 1 racing, 2024)

The chosen telemetry metrics in this case is Speed, Position, Throttle, DRS and Lap time Delta. For this analysis, the subplot matplotlib property was used. In order to have direct comparisons amongst all for telemetry. By four, it means there will be four graphs, Speed, Throttle, DRS and Lap time delta all plotted against position. This way one can compare and associate these telemetry metrics directly as they contain the same corresponding x values.

For the case study 1, this following was the output,

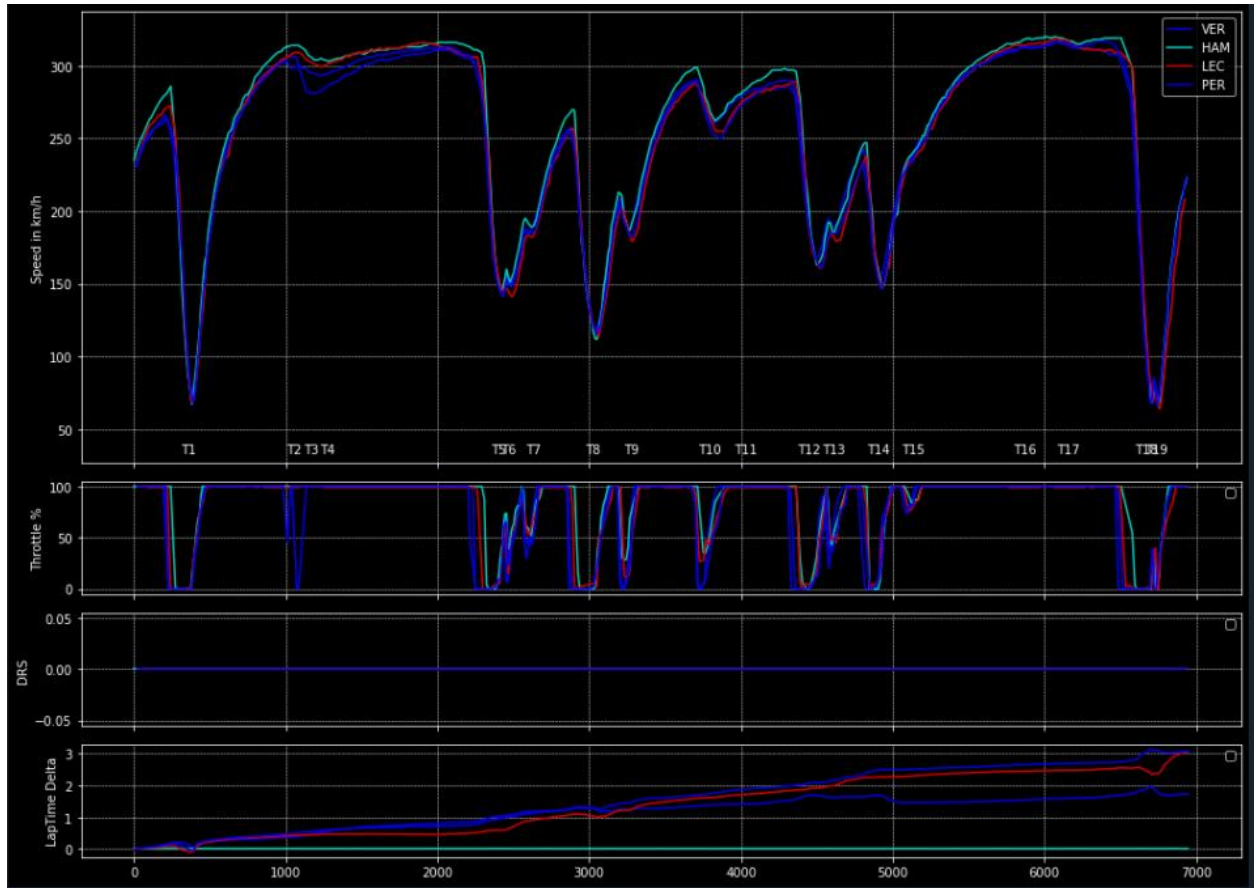


FIGURE 8: Belgium GP Telemetry Analysis Charts (Baraka, 2024).

For telemetry analysis in the year 2023, the focus was narrowed down to the main performers, namely Verstappen (VER), Hamilton (HAM), Leclerc (LEC), and Perez (PER). By limiting the number of plots on a single graph, it becomes easier and more effective to compare their performance.

In the speed subplot, when an individual plot is higher compared to the others, it indicates that the driver is moving at a relatively higher speed at that particular position. For example, if Hamilton's (HAM) green line is higher than Verstappen's (VER) blue line, it suggests that Hamilton's lap is faster.

The throttle percentage subplot simply illustrates the fluctuations from 0 to 100% in pressure applied to the throttle. Fluctuations in the DRS subplot typically reflect the usage of the Drag Reduction System (DRS) throughout the lap, indicating whether it is activated or deactivated at various points.

Regarding the Lap Time Delta subplot, the main compared lap time is set as the reference point at 0, which in this instance represents the Pole lap (the fastest lap). Any positive values plotted above 0 indicate slower laps compared to the reference lap, while negative values below 0 represent faster laps compared to the reference lap.

Code Implementation.

In Figure 8 to achieve this figure data is extracted similarly as in Figure 7,

- Lap data is extracted for each of the four drivers chosen, however, this time we will add position data to the main laps but utilizing a built in func-

```
fastest_lap = session.laps.pick_drivers(driver_id).pick_fastest()
car_data = fastest_lap.get_car_data().add_distance()
circuit_info = session.get_circuit_info()
```

tion.

- Now, the car_data object includes the "distance" property, providing access to corner data. Continuous variables such as speed, throttle, DRS, and delta can utilize this distance information to generate comprehensive illustrations.
- Additionally, grids incorporated to enhance visual appeal and create a more polished presentation.
- The lap time delta is synthesized an external fast f1 tool, utils. Utils has a delta-time function which takes two laps and compares them. Producing a delta lap made by finding the difference of the two.
- The main only one lap, the fastest by Hamilton (HAM) was used as the pole_lap and found difference with the rest of the laps.
- Utilizing subplots, four individual graphs are plotted against distance. A single label at the top succinctly details the names of the drivers, simplifying the graph interpretation.

5.2.1 Telemetry Evaluation

Upon closer examination, a significant pattern emerges, particularly within the Red Bull Racing team. Specifically, around corners T2, T3, and T4 (Eau Rouge). There is a conspicuous decrease in speed, accompanied by a sharp drop in throttle percentage by the Red Bull drivers (blue). This contrasts with the behaviour of other drivers, who maintain maximum throttle and maintain high speeds through these corners.

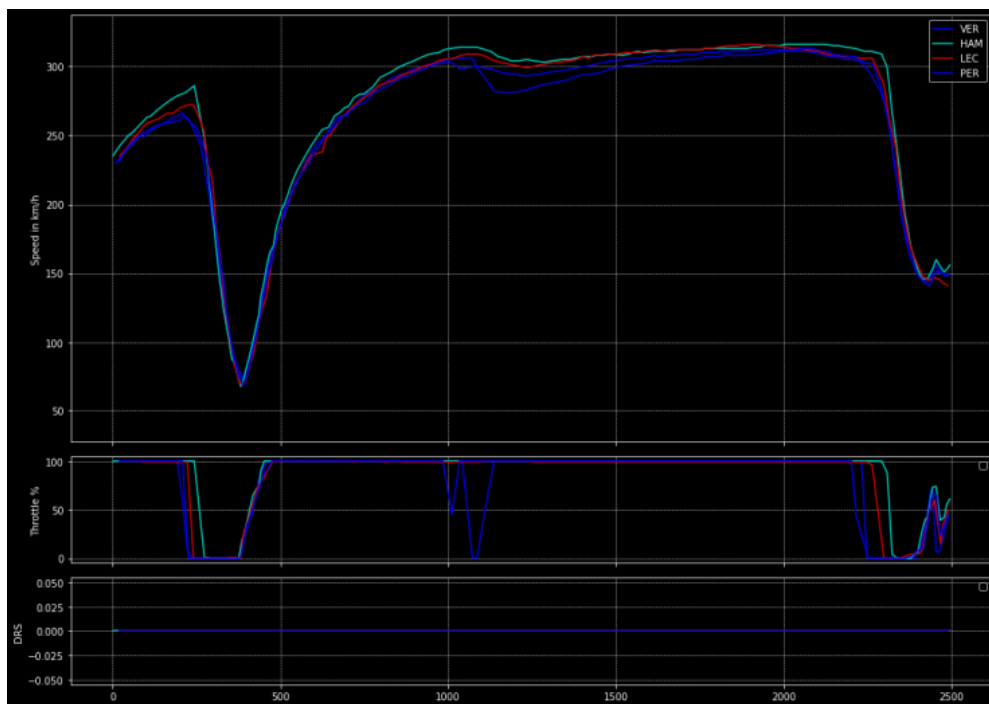


FIGURE 9: Close up sections from T1 to T7. (Baraka, 2024).

Analyzing the throttle data across the teams further corroborates this observation, revealing that the Red Bull Racing team completely disengages the throttle at approximately the same corners where the notable decrease in speed occurs.

This consistent action suggests a deliberate maneuver aimed at intentionally reducing speed at these specific corners. Furthermore, DRS was not used in these given laps as it is limited in its usage. At this point, the telemetry data provides us with valuable insights that can significantly influence our analysis of the case study.

Further investigation was required to determine whether this behavior was deliberate or incidental. By conducting an analysis on qualifying rounds of the race week for the Red Bull teams, the telemetry data gave a different story.

Upon further examination, in Figure 10, it appears that around Eau Rouge, during qualifying rounds, the Red Bull team maintains maximum throttle and achieves flat-out speed at this corner.

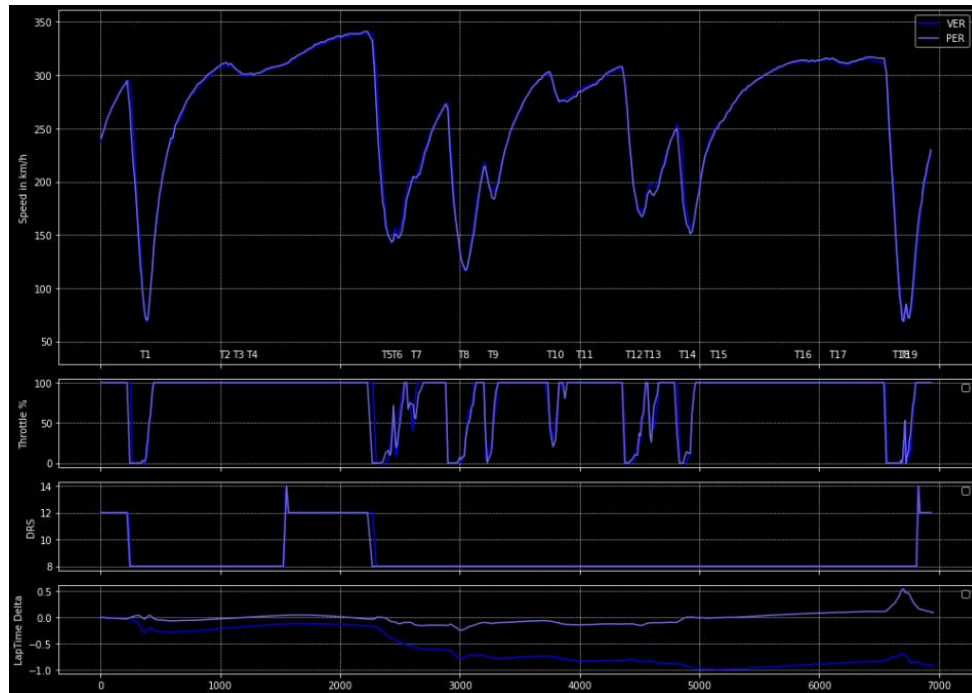


FIGURE 10: Red bull cars Qualifying Fastest Laps. (Baraka, 2024).

This observation suggests that Red Bull's actions were, at the very least, tactically deliberate, serving a specific purpose. Another notable difference is the active use of the DRS in portions of the lap. As this shift in throttle costs Red bull precious time at least 0.5s.

This suggests that the team discovered something that could negatively affect the car around this corner during qualifying. This is supported by the radio messages where they seemed to be insisting Max to restrain around this corner. (F1, 2023) Through a telemetry analysis, tactical insights and drivers' behavioral patterns emerge with clarity. This examination supports the theory that Hamilton's perceived performance boost was not the case. Instead, it suggests that Red Bull strategically opted for a cautious approach to circumvent potential penalties. This conclusion underscores the analytical depth and scientific rigor inherent in telemetry-driven investigations within Formula 1 racing.

6 CASE STUDY 2 ANALYSIS

In this analysis, attention is directed toward the meticulous examination of tire behavior and its impact on race outcomes. By focusing on specific laps and tire compounds, the aim is to discern patterns and trends in tire performance, shedding light on the challenges drivers face and the strategic decisions they make during the race. Through the lens of LapTime and degradation, key insights emerge, offering valuable insights into optimizing tire strategies and enhancing overall race performance.

In each analysis conducted, it is typically advisable to begin with a comprehensive data overview. However, a different approach was taken for tire performance analysis. Instead of a traditional data overview, a performance comparison graph, akin to Figure 7, was integrated into the tire performance and degradation graphs.

This innovative visualization technique facilitates a direct comparison among performance, degradation, and tire age, offering a holistic understanding of the intricate interplay between these variables.

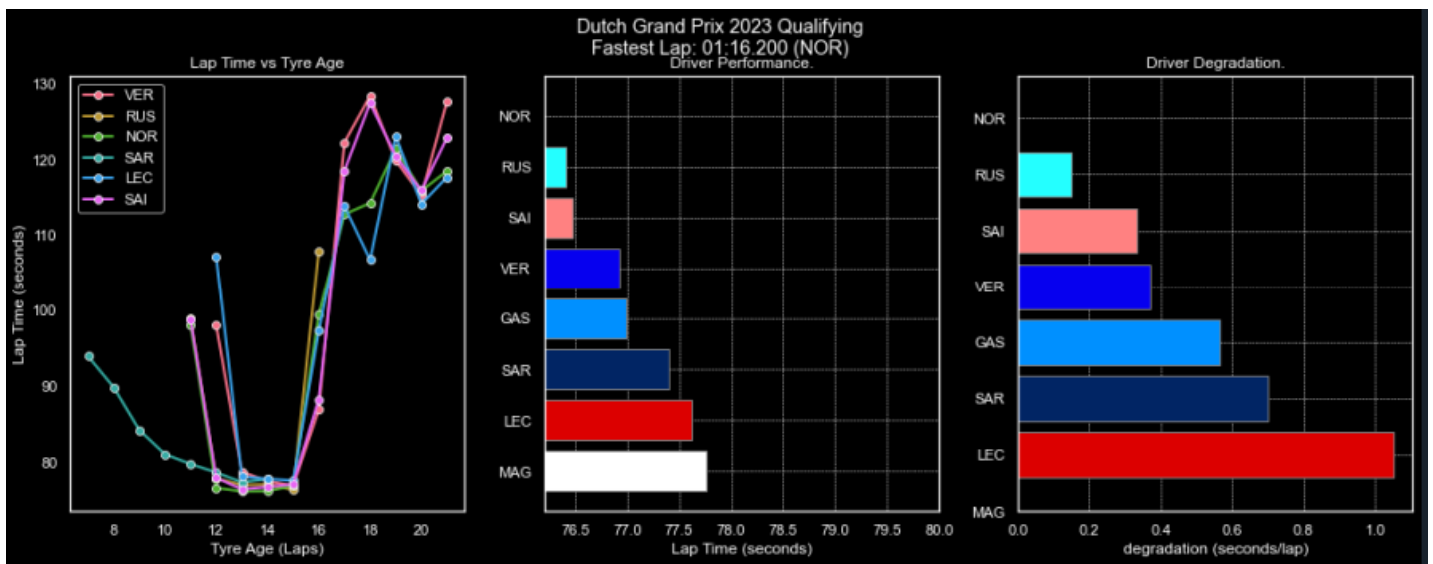


FIGURE 11: A graph of Tyre Age vs LapTime along with Driver performances and Degradation. (Baraka, 2024).

Code Implementation

- For the Driver Performance Graph, driver data is extracted and stored in a local instance variable of Lap class, with a function used to filter specific laps. Subsequently, the variable is further filtered to include only laps utilizing the 'SOFT' tire compound.
- Lap Times are sorted and plotted within the range of x values from 76 seconds to 80 seconds, offering a unique perspective compared to Figure 7.
- The Tyre Age Graph, positioned to the left, undergoes similar data extraction and filtering processes as the Driver Performance Graph.
- Lap Time is plotted against Lap number to showcase the performance trend with the 'SOFT' tire compound over the course of the race.

And for the Degradation Graph to the right,

- Similar procedures were followed for the construction of the following two graphs; however, additional processing steps were implemented.
- After data extraction, lap times for each driver were retrieved and processed through a looping mechanism. This process involved calculating the lap time delta (increase or decrease) for each consecutive lap. Subsequently, the mean of these lap time deltas was computed for each driver.
- The resulting data, along with the corresponding drivers, were sorted from lowest to highest values before being plotted on the graphs.

6.1 Data Validation

Since the same data was used, with the similar tables and calculations, the data validation method adopted here mirrors the approach outlined in the Fast F1 documentation, explained at section 5.1. This method entails comparing calculated timestamps with known accurate values and scrutinizing data consistency to ensure alignment with expected patterns. By leveraging these validation techniques, potential errors or inconsistencies in the data were identified and addressed, fortifying the reliability of the dataset for analysis.

6.2 Results

In the close FIGURE 12 below, a focused analysis was conducted by filtering all drivers except Verstappen (VER) and Norris (NOR). Interestingly, Norris (NOR) exhibited superior performance compared to Verstappen (VER) in the selected laps. However, FIGURE 10 reveals a contrasting trend, indicating that Norris (NOR) experienced significantly lower degradation compared to Verstappen (VER). This observation raises pertinent questions regarding potential factors contributing to this disparity, such as differences in car efficiency or tire composition.

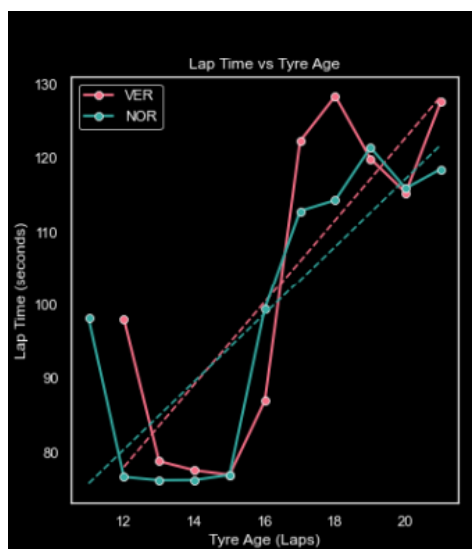


FIGURE 12: A close up with Trendlines of FIGURE 11

These insights underscore the importance of conducting thorough analyses with enhanced data quality, particularly across qualifying rounds, to glean valuable insights into performance dynamics and inform strategic decisions.

6.3 Evaluation of Tire Performance Analysis

The evaluation of the tire performance analysis underscores the close relationship between the three graphs, facilitating the extraction of critical performance-related insights encompassing both performance augmentation and degradation.

In the Tyre Age graph, an upward trend in the line signifies a decline in performance over time, while a downward trend indicates an improvement in perfor-

mance as the race progresses. The trendline serves as a comparative index between drivers, with a trendline positioned above another indicating that the corresponding driver performs less compared to their counterpart.

This visual representation offers valuable insights into the correlation between tire age and performance fluctuations throughout the race duration.

The degradation graph effectively illustrates the correlation between degradation values and the subsequent loss in performance. Higher degradation values are indicative of more pronounced performance decline, underscoring the significant impact of tire wear on driver performance during the race. Conversely, smaller degradation values denote minimal performance loss, emphasizing the critical role of strategic tire management in maximizing overall race performance. It's important to note that the rate of change of degradation plotted in relation to the smallest degradation provides valuable insights into the efficiency of tire utilization and its impact on overall race dynamics.

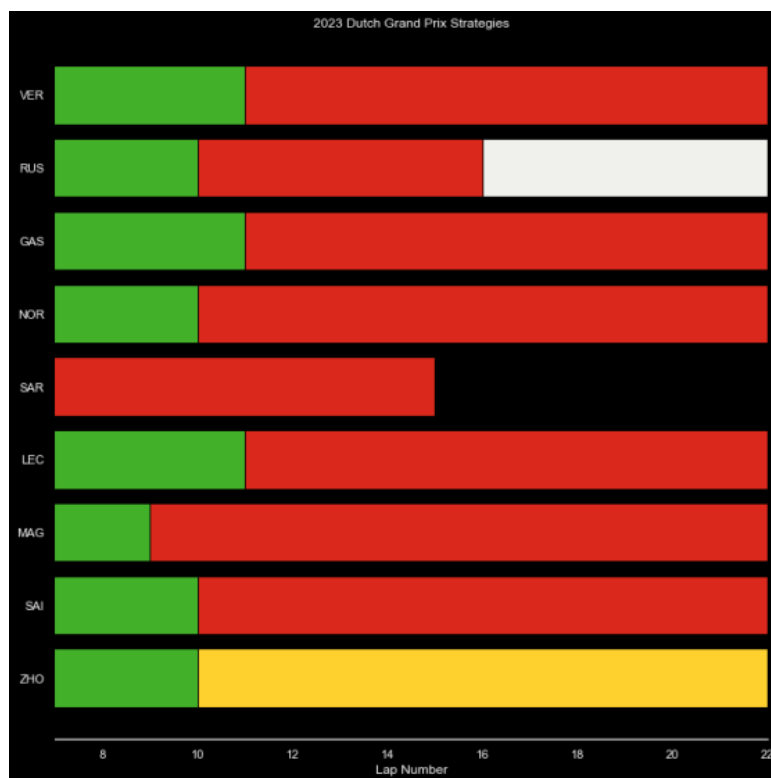


FIGURE 13: this graph shows the tires used for each lap. Green is Medium and Red is soft. (Baraka, 2024).

The selection of laps 7 to 22 was strategic, as this segment typically corresponds to the phase where most teams' drivers transition from medium to soft tires. Given the tendency for soft tires to experience quicker performance degradation, analyzing this interval offers valuable insights into the comparative performance dynamics between the two tire compounds.

FIGURE 13 represents all drivers and the tire strategies used for each laps. Green bar illustrates Medium tires, Red is soft, Yellow represents hard tires and White shows an early race exit.

FIGURE 12 was briefly implemented by,

- For the Driver Performance Graph, driver data is extracted and stored in a local instance variable of Lap class, and then Driver, Stint, Compound, LapNumber were extracted. Where the first three were used as a parameters to group the information in this variable.
- And then, the data is processed and plotted using width of each lap numbers covered by each compound.

Furthermore, this type of analysis is ideally suited for qualifying rounds, where teams aim to optimize their car's performance over a short duration. However, it's essential to acknowledge that qualifying data may not always provide clean and reliable insights due to various factors such as differing fuel loads, traffic, and varying utilization of the car's full performance potential. Despite these challenges, leveraging such analyses during qualifying sessions can offer valuable predictive capabilities for race performance and inform strategic adjustments accordingly.

7 CONCLUSION

This thesis explored Formula 1 analysis through two case studies, focusing on telemetry analysis in the Belgian GP and tire performance analysis in the Dutch GP. By utilizing the Fast F1 framework alongside the Anaconda framework, alongside tools like Matplotlib, the study sought to extract insights from processed data and present them through custom illustrations.

Case study 1 focused on telemetry analysis, offering comprehensive insights into driver performance, vehicle dynamics, and race conditions. By comparing these insights across different races, such as qualifying and race data, varying patterns and correlations emerge, providing valuable information with significant impacts on race outcomes. Through telemetry analysis, strategies and factors influencing race performance can be uncovered.

Case study 2 focused on tire performance analysis, provided insights into how each team's car setup affected performance while objectively examining tire strategy. It also assessed how car setups reacted to external factors like track conditions and weather, considering the varying responses of different tire compounds (soft, hard) to the track. These insights significantly impacted strategies for specific races and tracks, influencing decisions such as tire selection for different sets of laps and the timing of activating the DRS (Drag Reduction System) during the race. Compared to case study 1, tire performance analysis had a higher potential and impact on strategies. This analysis is to be conducted during qualifying to understand each team's car better and effectively plan for the main race.

The strength of this study is in the examination of various race analysis techniques and their impact on race strategies. By comparing these techniques across different races, the study provides a diverse and impactful perspective on how illustrations and analysis influence strategic decisions.

The clarity, complexity, and comprehensibility of the illustrations allow for the extraction of useful information that effectively addresses the analysis case study topic.

The weaknesses include challenges in implementing functions due to poor documentation and a scarcity of working examples, hindering effective use of the framework for complex illustrations. The lack of illustrations with more detailed and complex information limits the depth of analysis that could be achieved. The data validation process heavily depended on Fast F1's pre-existing validation features, as detailed in section 5.1, thereby constraining customization options for researchers. The study failed to accurately measure the influence of weather conditions on race performance, highlighting a notable gap in the analysis.

The report demonstrates that the analysis has yielded valuable observations regarding driver performance, race conditions, and the impact of external factors such as track conditions and weather on car setups and race strategies. Although the analysis's predictions may not possess a high level of accuracy, they still hold enough significance to facilitate informed decision-making that greatly influences the team's choices, strategies, and overall performance.

Future investigations should prioritize better planning and a deeper understanding of documentation to enhance the effectiveness of analysis methodologies. Augmenting more basic illustrations with advanced mathematical calculations, as demonstrated in Case Study 2, should also be explored, to enrich the analytical process. Future investigations should also explore the integration of weather data from Fast F1 and quantify its impact on other metrics such as telemetry and tire performance. While some experts downplay its significance, an analysis could quantify weather's impact on the metrics. Such insight could significantly enhance performance and sports competition.

While this thesis reports on the complexities of Formula 1 analysis, it serves as a stepping stone for future scholars to refine techniques, deepen insights, and unravel the hidden intricacies of motorsport analytics.

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